TA Session I: Notes

Microeconomics I, Part B — Choice under Uncertainty

Exercises 8, 4, 9, and 10 (in the order of the session)

Exercise 8 (I)

Data. Expected-utility decision maker. Possible occupations:

- A: working in the finance sector;
- B: working in the movie industry;
- C: working in the car industry.

Preferences: School 1 $\sim$ School 3 $\succ$ School 2.

(a) *Give a representation of the utility function.*

Each school corresponds to a lottery over (A, B, C):

$$L_1 = (0, 0, 1), \qquad L_2 = (1, 0, 0), \qquad L_3 = (0.9,\ 0.1,\ 0).$$

The decision maker's evaluations are

$$U(L_1) = u_C, \qquad U(L_2) = u_A, \qquad U(L_3) = 0.9\,u_A + 0.1\,u_B.$$

According to the text, the ranking of the lotteries is

$$U(L_1) = U(L_3) > U(L_2) \iff u_C = 0.9\,u_A + 0.1\,u_B > u_A.$$

Without loss of generality,

$$u_A = 0, \qquad u_B = x, \qquad u_C = \frac{x}{10}, \qquad \text{with } x > 0.$$

(b) *We assume that a new school (School 4) opens, with $L_4 = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$. How would the new school rank compared to Schools 1, 2, 3?*

$$U(L_4) = \frac{1}{3}\,u_C + \frac{1}{3}\,u_B + \frac{1}{3}\,u_A = \frac{1}{3}\cdot\frac{x}{10} + \frac{1}{3}\,x + \frac{1}{3}\cdot 0 = \frac{x}{30} + \frac{x}{3} = \frac{x + 10x}{30} = \frac{11}{30}\,x.$$

$$\boxed{U(L_1) = \frac{x}{10}, \qquad U(L_2) = 0, \qquad U(L_3) = \frac{x}{10}, \qquad U(L_4) = \frac{11}{30}\,x}$$

So

$$U(L_4) > U(L_1) = U(L_3) > U(L_2), \qquad \text{since} \qquad \frac{11}{30} > \frac{1}{10} \iff 11 > \frac{30}{10} = 3.$$

(c) *There are too many applications to School 4. The ministry of education decides that rather than applying to School 4 directly, the decision maker must apply to a lottery that leads to School 4 with probability α and to School 2 with probability $1 - \alpha$. For which values of α does the decision maker prefer to apply to School 1 rather than to the lottery that may lead to admission to School 4?*

The new procedure corresponds to the compound lottery $\alpha L_4 + (1-\alpha)L_2$. By linearity of expected utility,

$$U\big(\alpha L_4 + (1-\alpha)L_2\big) = \alpha\,U(L_4) + (1-\alpha)\,U(L_2) = \alpha\,\frac{11}{30}\,x + 0 = \frac{11}{30}\,\alpha x.$$

School 1 is preferred when

$$U(L_1) = \frac{x}{10} > \frac{11}{30}\,\alpha x \iff 3x > 11\,\alpha x \iff \boxed{\alpha < \frac{3}{11}} \qquad \text{(dividing by } x > 0\text{)}.$$

The lottery is practically all or nothing (School 4 or the least preferred school), so the person needs to be compensated more: α must exceed 3/11 before the lottery is worth applying to.

Exercise 4 (II)

Data. Consider an investor who has to choose between two assets:

- $x_A = (12, 6, 9)$, with equal probability;
- $x_B = (\bar{x}, \bar{x}, \bar{x})$: risk-free asset.

(a) *Assume $u(x) = \log(x)$. Find the certainty equivalent of x_A.*

The certainty equivalent c_A satisfies

$$\log(c_A) = \mathbb{E}[\log x_A] = \frac{1}{3}\big(\log 12 + \log 6 + \log 9\big) = \frac{1}{3}\log(648)$$

$$\iff \log(c_A) = \log\left(648^{1/3}\right) \iff \boxed{c_A = 648^{1/3} \approx 8.65.}$$

The expected payoff of the lottery, instead, is

$$\mathbb{E}(x_A) = \frac{1}{3}(12) + \frac{1}{3}(6) + \frac{1}{3}(9) = 4 + 2 + 3 = 9.$$

The monetary risk premium is

$$\rho_A = 9 - c_A \approx 0.35.$$

The asset pays 9 on average, but the agent is equally happy with 8.65 for sure. Those ≈ 0.35 are what the risk is costing, measured in monetary units.

(b) *Find the level of $\bar{x}$ such that the investor is indifferent between asset A and asset B (the risk-free asset).*

Indifference requires

$$\log(\bar{x}) = \mathbb{E}[\log x_A] \iff \boxed{\bar{x} = c_A = 648^{1/3} \approx 8.65.}$$

The agent prefers the risk-free asset for $\bar{x} > c_A$, and the risky asset for $\bar{x} < c_A$.

(c) *Assume now that $\bar{x} = \mathbb{E}(x_A)$. Find how many units h_A of asset A should be given to the agent to make him indifferent between h_A and one unit of the risk-free asset. Provide some verbal intuition.*

Here $\bar{x} = \mathbb{E}(x_A) = 9$ and $h_A > 0$. Indifference requires

$$\mathbb{E}\big[\log(h_A\, x_A)\big] = \log 9.$$

By the log rule, the left side is $\log(h_A) + \mathbb{E}[\log(x_A)] = \log(h_A\, c_A)$. Thus

$$\boxed{h_A = \frac{9}{c_A} = \frac{9}{648^{1/3}} \approx 1.04} \text{ units of the risky asset.}$$

About 1.04 units of the risky asset are needed to match one risk-free unit paying 9. One risky unit has a certainty equivalent below 9, so the agent needs extra units as compensation for bearing risk.

(d) *Take as reference point the expected payoff $\mathbb{E}(x_A) = 9$, around which the two extreme payoffs of asset A are symmetric, $12 = 9 + 3$ and $6 = 9 - 3$. Compute the probability premium $\pi(9, 3, u)$, that is, the π that solves*

$$u(9) = \left(\tfrac{1}{2} + \pi\right) u(12) + \left(\tfrac{1}{2} - \pi\right) u(6).$$

Compare it with the gap between $\mathbb{E}(x_A)$ and the certainty equivalent found in (a).

The probability premium solves

$$\log(9) = \left(\frac{1}{2} + \pi\right)\log(12) + \left(\frac{1}{2} - \pi\right)\log(6)$$

$$\iff \pi = \frac{\log(9) - \frac{1}{2}\big[\log(12) + \log(6)\big]}{\log(12) - \log(6)} = \frac{\log\left(9/\sqrt{72}\right)}{\log(2)} \approx \boxed{0.085.}$$

The probability of the higher payoff must rise from 0.5 to approximately 0.5850 (an increase of 8.496 percentage points), and the probability of the lower payoff falls to approximately 0.415.

Both $\pi > 0$ and $\rho_A = 9 - c_A > 0$ express aversion to risk: the agent needs either better odds or a monetary compensation to bear it. They are different measures for different lotteries here.

(e) *Compute the coefficient of absolute risk aversion.*

$$u'(x) = \frac{1}{x}, \qquad u''(x) = -\frac{1}{x^2}, \qquad \boxed{r_A(x) = -\frac{u''(x)}{u'(x)} = \frac{1}{x}.}$$

The investor has decreasing absolute risk aversion: a given additive risk is less costly to bear at higher wealth. At the reference payoff $x = 9$, $r_A(9) = 1/9$. Relative risk aversion is $x\, r_A(x) = 1$.

Exercise 9 (III)

Data. Expected-utility decision maker with $u(x) = x^a$, $0 < a < 1$.

(a) *Let X be a lottery whose outcome is uniformly distributed over $[0, 1]$. Determine $\mathbb{E}[u(X)]$.*

$$\mathbb{E}[u(X)] = \int_0^1 x^a \, dx = \left[\frac{x^{a+1}}{a+1}\right]_0^1 = \boxed{\frac{1}{a+1}.}$$

As X has a continuous distribution, an integral is used to compute the expected value.

(b) *Let Y be a lottery whose outcome is 0 with probability $\frac{1}{3}$ and 1 with probability $\frac{2}{3}$. Determine $\mathbb{E}[u(Y)]$.*

$$\mathbb{E}[u(Y)] = \frac{1}{3} \cdot 0^a + \frac{2}{3} \cdot 1^a = \boxed{\frac{2}{3}} \qquad \text{(all or nothing).}$$

(c) *Determine the value a^* (of a) for which the decision maker is indifferent between X and Y.*

To be indifferent, $\mathbb{E}[u(X)] = \mathbb{E}[u(Y)]$:

$$\frac{1}{a^*+1} = \frac{2}{3} \iff a^* + 1 = \frac{3}{2} \iff \boxed{a^* = \frac{1}{2}.}$$

(d) *If $a > a^*$, which lottery, X or Y, is preferred by the decision maker?*

We know that $\mathbb{E}[u(X)] = \dfrac{1}{a+1}$, with $0 < a < 1$.

- $X \sim Y$ if $a = \frac{1}{2}$.
- $X \succ Y$ if $0 < a < \frac{1}{2}$. For example, if $a = \frac{1}{4}$:

$$\mathbb{E}[u(X)] = \frac{1}{\frac{1}{4}+1} = \frac{1}{5/4} = \frac{4}{5} = 0.80 > \frac{2}{3}.$$

- $X \prec Y$ if $\frac{1}{2} < a < 1$. For example, if $a = \frac{3}{4}$:

$$\mathbb{E}[u(X)] = \frac{1}{\frac{3}{4}+1} = \frac{1}{7/4} = \frac{4}{7} \approx 0.57 < \frac{2}{3}.$$

For larger a, utility is less curved and the decision maker is less risk averse. Consequently, sufficiently risk-averse agents (small enough a) prefer X; less risk-averse agents prefer Y.

(e) *Determine, as a function of a, the coefficient of absolute risk aversion of the decision maker.*

For $x > 0$:

$$u'(x) = a\,x^{a-1}, \qquad u''(x) = a(a-1)\,x^{a-2},$$

$$r_A(x) = -\frac{u''(x)}{u'(x)} = -\frac{a(a-1)\,x^{a-2}}{a\,x^{a-1}} = -(a-1)\,x^{a-2-a+1} = -(a-1)\,x^{-1} = \boxed{\frac{1-a}{x}.}$$

Absolute risk aversion decreases both with wealth x and with the parameter a. The Arrow–Pratt derivative formula is not defined at $x = 0$, although utility at zero is well defined.

Exercise 10 (IV)

Data. A decision maker has preferences over lotteries represented by an expected utility with Bernoulli utility $u(x)$. Consider the lottery that pays 0, 4, and 9 with probabilities $\frac{1}{6}$, $\frac{1}{2}$, and $\frac{1}{3}$:

$$L = \left\{ (0, 4, 9); \left(\frac{1}{6}, \frac{1}{2}, \frac{1}{3} \right) \right\}.$$

(a) *Provide the definition of the certainty equivalent of L given u.*

For an increasing continuous Bernoulli utility u, the certainty equivalent c solves

$$u(c) = \mathbb{E}[u(X)] = \frac{1}{6}\,u(0) + \frac{1}{2}\,u(4) + \frac{1}{3}\,u(9).$$

Then

$$\boxed{c = u^{-1}\big(\mathbb{E}[u(X)]\big).}$$

The agent is indifferent between receiving c for sure and receiving the lottery, which has $\mathbb{E}(X) = 5$.

(b) *Compute the certainty equivalent of L given $u(x) = \sqrt{x}$.*

$$\mathbb{E}\big(\sqrt{X}\big) = \frac{1}{6}\sqrt{0} + \frac{1}{2}\sqrt{4} + \frac{1}{3}\sqrt{9} = 0 + \frac{1}{2}(2) + \frac{1}{3}(3) = 2.$$

Then

$$c = u^{-1}\big[\mathbb{E}\big(\sqrt{X}\big)\big] = u^{-1}(2) = 2^2 = \boxed{4.}$$

(c) *Compute the coefficient of absolute risk aversion given u and x.*

In general, $r_A(x) = -\dfrac{u''(x)}{u'(x)}$. For square-root utility and $x > 0$:

$$u'(x) = \frac{1}{2\sqrt{x}}, \qquad u''(x) = -\frac{1}{4x^{3/2}}, \qquad \boxed{r_A(x) = \frac{1}{2x}.}$$

It decreases with wealth. The derivative formula is not defined at $x = 0$.

(d) *Assume that the agent starts with $w = 12$ and owns the lottery. Find the minimum price he is willing to accept to sell the lottery.*

Keeping the lottery gives wealth $12 + X$ (the payoff associated with the lottery adds to w); selling it at price p gives certain wealth $12 + p$. The minimum price solves

$$\sqrt{12 + p} = \frac{1}{6}\sqrt{12 + 0} + \frac{1}{2}\sqrt{12 + 4} + \frac{1}{3}\sqrt{12 + 9} = 2 + \frac{\sqrt{3} + \sqrt{21}}{3},$$

$$p_{\min} = \left(2 + \frac{\sqrt{3} + \sqrt{21}}{3} \right)^2 - 12 \approx \boxed{4.85.}$$

Utility increases with the sale price, so selling is weakly preferred exactly when $p \geq p_{\min}$, i.e. $p \geq 4.85$. The certainty equivalent of total wealth is $12 + p_{\min} \approx 16.85$.

(e) *Compare the minimum price found above and the certainty equivalent, and provide an intuition of their difference based on the coefficient of absolute risk aversion.*

We obtained

$$c = 4 < p_{\min} \approx 4.85 < \mathbb{E}(X) = 5.$$

Since $r_A(x) = \dfrac{1}{2x}$ decreases with wealth, the agent requires a smaller premium to bear the same additive risk at higher wealth. The risk premium falls from $1 = 5 - c$ to $0.15 = 5 - p_{\min}$, where $5 = \mathbb{E}(X)$.